Understanding Fitch's Empirical Approach to Basel II Correlation Values

1. Context of the whole report

The report studies whether the **Basel II asset correlation assumptions** used in the IRB credit risk formula are empirically reasonable.

In Basel II, banks calculate regulatory capital for credit risk using inputs such as:

$$PD, \quad LGD, \quad EAD, \quad \rho.$$

Here:

$$PD = \text{probability of default},$$

$$LGD = \text{loss given default},$$

$$\rho = \text{asset correlation}.$$

The important issue is that **Basel II fixes the asset correlation parameter** ρ by regulation. Banks estimate PD and LGD, but they do not freely estimate ρ for regulatory capital purposes.

The Fitch report asks:

> Are Basel II's fixed correlation assumptions too high, too low, or broadly consistent with historical loss experience?

To answer this, Fitch uses historical loss-rate data and tries to infer the asset correlation that would make the Basel II formula reproduce the observed historical loss volatility.

The report explains that historical loss data are used to estimate the mean and volatility of losses. These are then translated into expected loss and unexpected loss, and finally into an implied asset correlation. (Fitch report, pages 5–7.)

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2. Summary of page 7

Page 7 gives the **technical method** for deriving empirically implied correlations.

The procedure is:

$$\text{historical loss rates} \rightarrow \mu, \sigma \rightarrow \text{fitted loss distribution} \rightarrow 99.9\% \text{ loss quantile} \rightarrow UL \rightarrow \rho.$$

The page says that Fitch fits several distributions to the loss-rate data and finds that the **beta distribution** gives the best fit. The beta distribution is useful because loss rates are naturally bounded between 0 and 1.

The beta distribution has two parameters:

$$\alpha, \beta.$$

These parameters are obtained from:

$$\mu = \text{mean loss rate},$$

$$\sigma = \text{standard deviation of loss rate}.$$

Once the beta distribution is fitted, Fitch calculates the **99.9% total loss**. This is consistent with Basel II, which uses a 99.9% confidence level for credit risk capital.

Then:

$$UL_{99.9\%} = 99.9\%\text{total loss} - EL.$$

Since:

$$EL = \mu,$$

we have:

$$UL_{99.9\%} = q_{99.9\%} - \mu.$$

Finally, Fitch finds the asset correlation ρ that makes the Basel II IRB formula produce the same UL.

So page 7 is essentially the **bridge between empirical data and Basel II theory**.

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3. Summary of page 8

Page 8 presents the results for three asset classes:

1. **Credit cards**
2. **Residential mortgages**
3. **Consumer lending**

For each asset class, the table reports:

$$EL, \quad \sigma, \quad LGD, \quad PD, \quad UL, \quad \rho,$$

and compares the empirically implied ρ with the Basel II regulatory correlation assumption.

The main message is:

For most asset classes, Fitch's empirically implied correlations are lower than Basel II's regulatory correlation assumptions.

For example:

Asset class	Empirical ρ	Basel II assumption
Credit cards, Federal Reserve data	1.32%	4.00%
Residential mortgages, Federal Reserve data	2.07%	15.00%
Consumer lending, Federal Reserve data	1.31%	9.19%

This suggests that Basel II is generally conservative relative to historical loss experience.

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4. Step-by-step mathematical explanation

Step 1: Start from historical loss rates

Suppose we have a time series of historical annual loss rates:

$$L_1, L_2, \dots, L_T.$$

For example, for credit cards, Fitch uses loss-rate observations from 1985–2007.

From this time series, compute the sample mean:

$$\mu = \frac{1}{T}\sum_{t=1}^{T} L_t.$$

This is interpreted as the **expected loss rate**:

$$EL = \mu.$$

Then compute the standard deviation:

$$\sigma = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(L_t - \mu)^2}.$$

This captures the historical volatility of loss rates.

In the table, these are the first two numerical rows:

$$EL(\mu), \quad \sigma.$$

For example, for **credit cards, Federal Reserve data**:

$$EL = 4.23\%,$$

$$\sigma = 1.02\%.$$

Step 2: Decompose expected loss into PD and LGD

In credit risk, expected loss is:

$$EL = PD \times LGD.$$

Fitch observes the historical average loss rate, so Fitch knows EL. However, historical loss-rate data do not automatically give a separate PD and LGD.

Therefore, Fitch assumes an LGD and backs out PD:

$$PD = \frac{EL}{LGD}.$$

For credit cards, Federal Reserve data:

$$EL = 4.23\%,$$

$$LGD = 71.60\%.$$

Therefore:

$$PD = \frac{4.23\%}{71.60\%} = 0.0591 = 5.91\%.$$

The table reports:

$$PD = 5.90\%.$$

So this row is obtained directly from:

$$PD = \frac{EL}{LGD}.$$

The LGD values are assumed for the purpose of decomposing EL into PD and LGD; Fitch notes that these PD and LGD values are illustrative and should not be interpreted as Fitch's own average PD and LGD estimates for the asset class. (Fitch report, page 8.)

Step 3: Fit a beta distribution to losses

Fitch assumes the loss rate L follows a beta distribution:

$$L \sim \text{Beta}(\alpha, \beta).$$

The beta distribution is defined on:

$$0 \leq L \leq 1.$$

This is appropriate because a loss rate is a percentage.

For a beta distribution, the mean is:

$$\mu = \frac{\alpha}{\alpha + \beta}.$$

The variance is:

$$\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

Page 7 gives the rearranged formulas:

$$\alpha = \mu\left(\frac{\mu(1-\mu)}{\sigma^2} - 1\right),$$

$$\beta = (1-\mu)\left(\frac{\mu(1-\mu)}{\sigma^2} - 1\right).$$

So once μ and σ are known, α and β are determined.

For the credit-card example:

$$\mu = 0.0423,$$

$$\sigma = 0.0102.$$

Then:

$$\alpha = 0.0423\left(\frac{0.0423(1 - 0.0423)}{0.0102^2} - 1\right) \approx 16.43.$$

And:

$$\beta = (1 - 0.0423)\left(\frac{0.0423(1 - 0.0423)}{0.0102^2} - 1\right) \approx 371.95.$$

So the fitted loss distribution is approximately:

$$L \sim \mathrm{Beta}(16.43,371.95).$$

Step 4: Find the 99.9% loss quantile

The 99.9% total loss is the number $q_{0.999}$ such that:

$$P(L \leq q_{0.999}) = 0.999.$$

Equivalently:

$$q_{0.999} = F_L^{-1}(0.999),$$

where F_L^{-1} is the inverse cumulative distribution function of the fitted beta distribution.

For the credit-card example, this gives approximately:

$$q_{0.999} \approx 8.03\%.$$

This is not yet unexpected loss. It is the **total loss at the 99.9% confidence level**.

Step 5: Convert total loss into unexpected loss

Unexpected loss is the difference between the high quantile loss and expected loss:

$$UL_{99.9\%} = q_{0.999} - EL.$$

For the credit-card example:

$$UL_{99.9\%} = 8.03\% - 4.23\% = 3.80\%.$$

This gives the table entry:

$$UL = 3.80\%.$$

So the table's UL row comes from the beta distribution quantile minus the mean loss rate.

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5. How the Basel II formula is used

Now Fitch has:

$$PD, \quad LGD, \quad UL.$$

The remaining question is:

> What asset correlation ρ would make the Basel II formula generate this same UL?

For retail exposures, the Basel II IRB formula on page 7 can be written as:

$$K = LGD \cdot \Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{\rho}\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) - PD \cdot LGD.$$

Here:

$$K = UL.$$

So Fitch solves:

$$UL_{\text{Basel}}(\rho) = UL_{\text{empirical}}.$$

That is:

$$LGD \cdot \Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{\rho}\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) - PD \cdot LGD = UL_{\text{empirical}}.$$

For the credit-card example:

$$PD = 5.90\%,$$

$$LGD = 71.60\%,$$

$$UL_{\text{empirical}} = 3.80\%.$$

So Fitch solves:

$$0.716 \cdot \Phi\left(\frac{\Phi^{-1}(0.0590) + \sqrt{\rho}\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) - 0.0590 \cdot 0.716 = 0.0380.$$

The solution is:

$$\rho \approx 0.0132.$$

In percentage form:

$$\rho = 1.32\%.$$

That is the table entry.

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6. How each row of the table is connected

The rows of the table follow this sequence:

$$\boxed{\text{Historical loss data}}$$

gives:

$$\boxed{EL = \mu}$$

and:

$$\boxed{\sigma}$$

Then:

$$\boxed{PD = \frac{EL}{LGD}}$$

Then μ and σ determine the beta distribution:

$$\boxed{\alpha, \beta}$$

The beta distribution gives:

$$\boxed{q_{0.999}}$$

Then:

$$\boxed{UL = q_{0.999} - EL}$$

Finally, solve the Basel II formula for:

$$\boxed{\rho}$$

So the table is not a collection of independent numbers. The dependencies are:

$$EL, \sigma \to \text{beta distribution} \to UL,$$

$$EL, LGD \to PD,$$

$$PD, LGD, UL \to \rho.$$

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7. Worked example: Credit cards, Federal Reserve data

The page 8 table gives:

$$EL = 4.23\%, \quad \sigma = 1.02\%, \quad LGD = 71.60\%.$$

Step 1: PD

$$PD = \frac{4.23}{71.60} = 5.90\%.$$

Step 2: Beta parameters

$$\mu = 0.0423, \quad \sigma = 0.0102.$$

$$\alpha \approx 16.43, \quad \beta \approx 371.95.$$

Step 3: 99.9% total loss

$$q_{0.999} \approx 8.03\%.$$

Step 4: Unexpected loss

$$UL = 8.03\% - 4.23\% = 3.80\%.$$

Step 5: Implied asset correlation

Solve the Basel II formula:

$$UL_{\text{Basel}}(\rho) = 3.80\%.$$

This gives:

$$\rho = 1.32\%.$$

Therefore the first column of the table is obtained as:

Row	Value	Explanation
EL (μ)	4.23%	Mean historical loss rate
σ	1.02%	Standard deviation of historical loss rate
LGD	71.60%	Assumed LGD
PD	5.90%	4.23/71.60
UL	3.80%	99.9% beta quantile minus EL
ρ	1.32%	Basel II implied correlation solving for same UL
Basel II assumption	4.00%	Regulatory fixed value

8. Interpretation of the results

The main interpretation is:

$$\rho_{\text{empirical}} < \rho_{\text{Basel II}}$$

for many asset classes.

This means that Basel II often assumes stronger systematic dependence than Fitch estimates from historical losses.

For example:

$$\rho_{\text{credit card empirical}} = 1.32\%,$$

while:

$$\rho_{\text{Basel II}} = 4.00\%.$$

So Basel II gives a higher capital requirement than the historical loss data alone would imply.

But Fitch does not conclude that Basel II is simply “too conservative.” The report emphasizes that historical data may understate crisis-period dependence. During financial stress, defaults can cluster, loss volatility can rise, and correlations can jump. Therefore, the higher Basel II correlation can be interpreted as a prudential buffer against stress, model error, and differences across banks.